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Improving the cross-cultural functioning of deep artificial neural networks through machine enculturation

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ABSTRACT

Artificial intelligence applications are being rapidly deployed around the world, where they need to interact with humans exhibiting different sociocultural values and related behaviors. Through machine enculturation, computers are supposed to assimilate these values so that they can better relate to humans. But artificial intelligence research has not yet fully managed that challenge. This article uses a deep artificial neural network to mimic the functioning of the human emotional brain by relating value priorities, opinions, and other factors with subjective well-being. It highlights that a network's hyperparameters configured to successfully train and perform with data from one country may not necessarily train and perform well with data from another country. Advancing our understanding of machine enculturation, the analysis demonstrates that the network's performance can be improved by pooling data across countries and coding binary country placeholder variables into the input vector.

Introduction

With the rapid deployment of artificial intelligence (AI) applications around the world, "we are entering an age in which machines are tasked not only to promote well-being and minimize harm, but also to distribute the well-being they create, and the harm they cannot eliminate" (Awad et al., 2018, p. 59). The classic example is that of an autonomous vehicle that is about to crash and cannot find a path to save everyone. Moreover, this vehicle's AI system might have been trained and tested with data from one country where people think and behave in a certain way. When the vehicle is deployed in another country, will the human actors behave in a similar way? Does the vehicle's system have to be retrained and tested again? Or can social norms, customs, and values be instilled into computers so that they can more readily relate to humans and avoid creating physical or psychological harm?

Take another example into consideration, managers across all levels spend more than half of their time on administrative coordination and control. Today's AI systems are already automating some of these tasks, and future systems are likely to assist in decision making, data-driven simulations, and discovery activities. In fact, a majority of managers believe that they will soon trust the advice of intelligent systems in making business decisions (survey by Accenture; Kolbjørnsrud et al., 2016). But what social skills will an AI system need so that managers can interact with it in a genuine way?

Across big data analytics, a common challenge is to handle the scale of the dataset and the geographical distribution of interacting systems. Machine learning is increasingly applied for data mining, so that hidden knowledge within these huge datasets can be extracted (Esposito et al., 2019). A non-marginal aspect contributing to such knowledge is certainly cultural information. How can geography and culture be embedded into AI systems?

While in natural sciences broad discussions have taken place about the role of sociocultural factors in humans, social skills have long been ignored in research dealing with the construction of intelligent artifacts, namely AI systems and robotics (Cassell, 2019; Dautenhahn, 1995). AI is today a thriving field (Nah et al., 2020) and is thought by some to be the future of our society (Verma et al., 2021). However, AI is still hardly used in an explicit cross-cultural context (Messner, 2022a; van Tulder et al., 2019). "The mere phrase 'soft sciences' is telling as to the status of social science in computer science" (Cassell, 2019, p. 30). It seems that an examination of the correct cross-cultural functioning of AI algorithms has been neglected so far, but a similar learning behavior of intelligent artifacts across countries would be crucial for a successful deployment around the world.

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Academics and practitioners agree that technology is not the main challenge to adopting AI, but the main impediments are, instead, cultural hurdles, processes, and personal value priorities (Borges et al., 2021; Duan et al., 2019; Kitsios & Kamariotou, 2021). Consequently, it has recently been proposed to use machine enculturation to instill sociocultural values in AI systems (Riedl & Harrison, 2016; Robinson, 2020), because "robotics that act in accordance with social values will naturally avoid situations where humans will be harmed or inconvenienced" (Riedl, 2016, p. 3). Early on it was noted that "such a competence might be much more important for the acceptance and 'social integration' of the robots into human societies than a robot's outer appearance" (Dautenhahn, 1995, p. 334). The problem is that intelligent artifacts are created by people with their own set of sociocultural values, judgements, moralities, ethics, and other biases (Dwivedi et al., 2021). Because "models are opinions embedded in mathematics" (O'Neil, 2016, p. 21), intelligent artifacts still use human culture as a source for their relationship blueprint.

Machine learning is not a magic tool for software developers. Instead, different machine learning algorithms need to be used for different problems and acceptable execution times. Suitable input features need to be constructed to build platforms capable of making intelligent decisions (Lee et al., 2019). Although sociocultural values have been identified as important variables, they have been ignored in AI thus far. This present research is a first step in closing this gap.

Specifically, I use a deep artificial neural network to understand how a person's subjective well-being depends on, inter alia, personal value priorities and opinions held by this person (Mesquita et al., 2016; Shweder et al., 2008). Well-being consists of a cognitive and affective component (Aknin et al., 2013; Diener, 2000; van der Meer & Wielers, 2013). It is an emotional mental state created out of the combination of basic psychological operations and ingredients, which can be mapped to intrinsic networks in the human brain (Lindquist & Barrett, 2012). The formation of well-being is not only related to a person's unique personality, but also to cultural norms and practices at the societal level (Diener et al., 2000; King et al., 2004; Sortheix & Lönnqvist, 2014). Using 422,985 survey data from 32 European countries and Israel from the first nine rounds of the European Social Survey (ESS; ESS 1-9, 2020), I optimize the configuration and hyperparameters of a deep artificial neural network for one country (Germany). I then use the same configuration and hyperparameters to train separate networks with data from further countries. For Germany and, indeed, many of the 33 countries, the loss function is satisfactory both with respect to the networks' interpolation and generalization capabilities. Yet, for some countries, the networks do not interpolate quite that well, or only interpolate well without an adequate generalization capability.

For machine learning to create business value through strategic use, the above constitutes a typical practical implementation challenge (Kitsios & Kamariotou, 2021). Using a simple form of machine enculturation as a possible way forward, I propose a pooled solution, which includes country information in the input vector and trains only one network with data from all 33 countries. This yields a better overall interpolation and generalization capability compared to training a separate network for each country.

In a nutshell, the research presented in this article aims to improve learning behavior of deep artificial neural networks across countries and cultures. It contributes to the so far limited methodological toolbox of machine enculturation, and the proposed methodology can be used by anyone deploying AI artifacts around the world.

Machine learning

Overview

AI with its trinity of data, algorithms, and computing infrastructure has seen tremendous progress over the last decade (Ekman, 2022). One approach to AI is machine learning, that is, using an algorithm to feed raw data into a flexible, dynamic, and nonlinear model (Sarle, 1994). At first, such an artificial neural network had only one hidden layer, which limited its ability to form abstractions. Currently, deep learning allows for design of neural network models composed of multiple hidden processing layers; algorithms progressively learn complicated concepts by extracting higher-level features from raw input data (e.g., Goodfellow et al., 2016; LeCun et al., 2015). Computing scientists only managed to build first experimental versions in 2006, followed by commercially viable applications in 2010 (Efron & Hastie, 2016, p. 364; Glorot & Bengio, 2010; Goodfellow et al., 2016, p. 23; G. E. Hinton et al., 2006). Even though deep learning algorithms are ideal for handling individual-level data from a heterogeneous sample, they are mathematically not designed to directly address different social norms, customs, and values.

However, AI assumes that machines are ultimately capable of making better decisions than humans, and "there is a huge potential to use these methods across different problem domains" (Kar, 2016, p. 30), including supply chain, information systems, and production management. Cognitive computing goes one step further and enables AI to interact with individuals in a human-like way, thereby interpreting contextual meaning. Motivated by the capability of the human mind, cognitive computing captures human thought processes and has the ability to improvise (Gupta et al., 2018; Modha et al., 2011). The "ultimate goal is a unified theory of human cognition" (Newell, 1990, p. 111).

In recent machine learning literature, there are three streams of research that address human emotions. First, the discipline of affective computing relates to emotion recognition and expression (Ehsan Lotfi & Akbarzadeh-T., 2014) by linking physiological signals and emotions, selecting stimuli for certain emotional states, extracting emotion-characteristic features, understanding the mechanistic causality of emotions, and recognizing emotions by fusing multi-modal information (Zhang et al., 2020). Second, emotions are modeled as a biological process via connectionist approaches for neuropsychological issues (e.g., Frewen et al., 2008; Grossberg & Seidman, 2006; Levine, 2007). Third, the anatomic brain is taken as a model for a computational learning process (Lotfi et al., 2018; e.g., Lotfi & Akbarzadeh-T., 2014). For this research, I follow the second stream and build a computerized model, which captures the human thought process and includes personal value priorities.

Machine enculturation

If we fail to understand sociocultural diversity when creating AI systems, it may negatively impact equal access to the advantages of technology (Dignum, 2019, p. 101; Robinson, 2020). But a core principle of the Universal Declaration of Human Rights (UDHR) states that "all human beings are born free and equal in dignity and rights" (UN, 1948, p. 72). And so, there is currently "growing interest in the effects of intelligent machines on human dignity" (Dignum, 2019, p. 15), in particular considering the healthcare, insurance, and law domains (e.g., Rawat et al., 2021).

Machine enculturation is one method to ensure that sociocultural values are present in computers so that they can more readily relate to humans and avoid social disruptions or even physical harm (Riedl, 2016; Riedl & Harrison, 2016; Robinson, 2020). Enculturation is the result of socialization experiences of the youths within their communities; it leads to the formation of the youths' social identity (Gonzales et al., 2005). Unfortunately, state of the art artificial intelligence research is not yet ready to tackle the problem of cultural value alignment (Riedl & Harrison, 2016).

Manually encoding sociocultural values into computers to recreate sociocultural practices is intractable (Riedl, 2016). Accordingly, it has been proposed that an artificial intelligence system can be made to learn tacitly held societal values from reading a corpus of stories representative of a society (Riedl & Harrison, 2016). Another proposed framework is mutual shaping and coproduction, which explores the interaction be-

Table 1
Sample characteristics.

Country	*N*	Female	Well-being	
			affective	cognitive
Austria	13222	53.804%	7.541	7.573
Belgium	16110	50.639%	7.738	7.439
Bulgaria	10522	57.023%	5.352	4.587
Croatia	4943	57.475%	6.963	6.530
Cyprus	5190	53.083%	7.397	7.096
Czech Republic	17614	52.316%	6.805	6.569
Denmark	12408	49.283%	8.327	8.461
Estonia	15314	57.549%	6.876	6.416
Finland	17955	51.406%	8.057	7.980
France	17061	53.971%	7.174	6.371
Germany	25700	49.467%	7.390	7.133
Greece	9759	56.000%	6.455	6.066
Hungary	14793	56.020%	6.357	5.736
Iceland	3072	50.586%	8.256	8.128
Ireland	20463	53.350%	7.459	7.126
Israel	14910	53.924%	7.598	7.322
Italy	7538	52.308%	6.980	6.834
Latvia	2898	63.975%	6.641	6.088
Lithuania	9993	62.224%	6.373	5.821
Luxembourg	3187	50.329%	7.833	7.808
Netherlands	16859	54.636%	7.786	7.654
Norway	14654	47.120%	7.990	7.868
Poland	15624	52.298%	7.075	6.736
Portugal	16043	59.509%	6.671	5.813
Russia	12458	59.672%	6.106	5.519
Slovakia	9874	56.289%	6.558	6.215
Slovenia	12232	53.663%	7.249	6.918
Spain	17169	50.836%	7.552	7.187
Sweden	15929	49.884%	7.867	7.866
Switzerland	15402	51.863%	8.062	8.058
Turkey	4272	54.260%	6.009	5.886
Ukraine	9987	62.621%	5.634	4.488
United Kingdom	19830	55.078%	7.483	7.147

This table shows the characteristics of the ESS sample. *N*: Number of respondents for each country. *Female*: Percentage of female respondents. *Well-being*: Country average of affective and cognitive well-being, scale of 0.10.

tween artificial intelligence systems and society by paying attention to how culture influences the way technologies are designed, used, and evaluated (Šabanović, 2010).

In this paper, I suggest and describe a rather primitive form of machine enculturation, namely adding the country as a placeholder variable to the input vector of a deep artificial neural network. I empirically show that this simple method already improves cultural alignment of deep learning algorithms because complex abstractions are learned throughout the layers of the network based on simpler abstractions formulated in a preceding level (Najafabadi et al., 2015).

Method

Following this brief introduction to machine learning and the necessity of machine enculturation of intelligent artifacts, I now describe the data, outline the research model, and explain the configuration and the training process of the deep artificial neural network. Explicitly, my aim is to show that simply adding country information as a placeholder variable improves the performance of a deep learning network.

Data

The European Social Survey (ESS 1-9, 2020) intends to measure cross-national attitudes in Europe. The first nine rounds of the ESS (2002 to 2018) contain 422,985 representative samples of adults residing in private households across 32 European countries and Israel (Table 1). Because the ESS uses identical (translated) items and response formats, the datasets are appropriate for evaluating questions of differences between countries (Broderick et al., 2007; Fitzgerald & Jowell, 2010).

The ESS contains two items for the affective and cognitive aspects of well-being, which are measured on a 11-point Likert-scale. I identify and select additional 159 items (Messner, 2022b), which capture respondents' personal value priorities, trust, health, and demographic factors. A detailed variable list is available in the Online Supplement.

Research model

By relating these 159 items about personal value priorities and other factors with affective and cognitive well-being, I attempt to mimic the functioning of the human emotional brain. My first neural network (Configuration A) is a classic setup with these 159 items in the input vector and the two well-being items in the output vector. In my second network (Configuration B), I add the survey respondent's country (as available in the ESS) as a binary placeholder variable to the input vector, resulting in 192 nodes.

Configuration of the deep artificial neural network

To prepare the data, I transform categorical variables into binary placeholder variables and rescale all other variables to an interval of [0..1]. I fill in missing values using the k-Nearest Neighbors approach ($k = 5$; points weighted by the inverse of their distance) or as a truncated random array based on the normal distribution in other countries.

Then, I configure a deep neural network to learn a vector regression problem with Keras 2.4.3, an open-source Python software library for deep learning and interface to Google's TensorFlow. Depending on the configuration, there are 159 or 192 in the input layer describing an individual survey respondent. The output layer features two nodes representing the person's well-being. Four hidden layers with 500 nodes each enable the network to hierarchically combine features at increasing levels of abstraction (Ekman, 2022, p. 163).

I use the ADAM (adaptive moment estimation) optimization algorithm (Kingma & Ba, 2015) together with a rectified linear unit (ReLU) activation function with He normal initialization (He et al., 2014) for the input and hidden layers. For the output layer, I choose a linear function with a Glorot uniform weight initialization (Glorot & Bengio, 2010). I add an L_2 regularizer with a very small weight decay parameter of $\lambda = 0.01$ (Kuhn & Johnson, 2016, p. 144) to each layer and the bias in order to improve the model's generalization capability for small datasets. I randomly remove a subset of neurons from the network with a dropout rate of 10% (Geoffrey E. Hinton et al., 2012) and impose a weight constraint of three to the remaining nodes after the dropout (Booz et al., 2018). Because the variables are all derived from answers to Likert-type survey questions, adding a zero-centered Gaussian Noise (standard deviation of 0.01) as random data augmentation to the input and output layers aids generalization and fault tolerance through embodying a smoothness assumption (Goodfellow et al., 2016, p. 241; Reed & Marks II, 1999, pp. 273–274). This documentation of the hyperparameter settings is important because they can affect the performance of deep artificial networks (Knoke & Wick, 2021).

Training the network

Using the ESS dataset for 33 countries, I now train my two network configurations A and B in three different ways. First, for each country, I train a separate network using configuration A. Second, and continuing with configuration A, I present the network with pooled data from all 33 countries. Third, I use configuration B, which has country information added to the input vector as a simple form of machine enculturation. I again train it with pooled data. In the following section, I describe the results of these three different ways.

Results

Several networks trained separately

For the ESS dataset from Germany (N = 25,700), the mean squared error loss function (Ekman, 2022, pp. 154–155) shows that the model rapidly converges and continuously improves both the training and generalization error without any sign of overfitting (Fig. 1). Using the same configuration and hyperparameters for other countries, the networks' performance shows differences. Fig. 1 provides examples of countries with large, medium, and small datasets. For the United Kingdom, both loss functions converge, but at a higher error level than for Germany. For Finland and the Czech Republic, both loss functions converge well, but show slight signs of overfitting after about 25 epochs. For other countries, the generalization error does not approximate the training error as well, even when using early stopping (e.g., Ireland, Spain, Switzerland, and Russia).

For Netherlands, Israel, and Italy, however, the training error takes more epochs before it begins to flatten out. Notwithstanding the longer training time, in Italy, the final training error is better than in Germany. However, the generalization error for these three countries shows early overfitting, that is, the networks' generalization capabilities worsen the more they are trained. The networks will not work well for prediction tasks in those countries.

To ensure that the above observations are not due to suboptimal local minima, I repeat the training process several times with different random seeds and reshuffled data. The results are comparable.

For countries with large and medium sample sizes (Germany, N = 25,700 to Italy, N = 7,538), sample size differences do not seem to be able to explain the above differences of network training behavior. However, countries with smaller sample sizes below N = 5,000 (Turkey, Croatia, Luxembourg, and Iceland in Fig. 1) all appear to show a very good interpolation and no overfitting, but do exhibit a poor generalization capability. This is a known effect (for a discussion of deep learning with small sample sizes see Koppe et al., 2021).

One plain network trained on pooled data

When training one plain network on data from all 33 countries (Fig. 2, Panel A), I remove the dropouts and reduce the L_2 regularizer to $\lambda = 0.001$. The training and generalization errors reach 0.626 and 0.630, respectively. This is a better performance than those achieved by any separately trained network.

One network trained on pooled data with machine enculturation

In addition to pooling all data together, I add country information as binary placeholder variables to the input vector. Other than the input vector now having 192 nodes, I use the same network configuration as before. The encultured network shows a further improved performance (Fig. 2, Panel B). Specifically, the training error falls below 0.619 and the generalization error is 0.620.

This rather clearly shows the benefit of a simple form of machine enculturation, such as adding the survey respondents' countries as binary placeholder variables to the input vector, for explaining and predicting human emotions with deep artificial neural networks.

Discussion

In the deep learning literature, a stream of research expresses human emotions as a biological process via connectionist approaches for neuropsychological issues (Frewen et al., 2008; Grossberg & Seidman, 2006; Levine, 2007). However, and to the best of my knowledge, this stream of literature has not yet brought aspects of different (national) cultural contexts into play. This is exactly the focus of the present article, in which I use a deep artificial neural network and value priorities, opinions, and other demographic factors in the input vector to mathematically mimic the functioning of the human emotional brain. Using data from the ESS, I investigate signals and patterns in the data (Denk et al., 2012; Thomas, 2012) and, through novel findings, contribute to the methodological advancement and practical application of deep learning across cultural contexts. Pertinently, my empirical experiments on the ESS dataset from 33 countries support the importance of machine enculturation. I have shown that an artificial intelligence system, encultured with something as simple as information about an individual's country, performs better than a plain system or systems trained separately for each country.

This is especially remarkable since the black-box brain of the neural network – and similarly those of most other present-day intelligent artifacts – does not exceed the level of insects. The network in configuration B features a total of 2,194 nodes and 847,000 weights, which is comparable to the 2,500 Kenyon cells in the brain of a common fruit fly (Heisenberg, 1998). The human brain, though, has about 120 billion neurons and 100 trillion synapses (Dance & McAllister, 2020; Herculano-Houzel, 2009) – a level which has only been surpassed in 2022 by BaGuaLu, a brain-scale AI model (Ma et al., 2022).

Limitations and future research

The finding that binary country information can already produce an enculturation effect is important, because, as illuminated by Riedl and Harrison (2016), artificial intelligence research is not yet ready to tackle the problem of value alignment in a more sophisticated manner. As with all research, this study has a few limitations, which quite naturally open up exciting avenues for further research.

First, I have used a dataset from the social sciences in this paper and I now encourage other researchers to replicate my findings with a variety of other datasets, perhaps even from areas where applications perform actual interactions with humans. Another example would be the alignment between machine learning tools and business strategy, where culture plays a crucial role (Duan et al., 2019; Kitsios & Kamariotou, 2021).

Second, the ESS covers 32 European countries and Israel. It would be interesting to see how the country-placeholder approach works in other intelligent artifacts trained with data containing further regions of the world.

Third, I have used only one clustering dimension in the input vector, namely binary country information. One could imagine other dimensions, such as information about one's mother tongue or religious affiliation. These would be added as further binary placeholder variables to the input vector. It is conceivable that this additional form of machine enculturation will further improve the network's training behavior.

Fourth, in any neural network, the self-organization of the hidden layers during the training process is opaque. How the placeholder variables for the countries contribute to achieving a better overall performance is not an insight understood by humans (Futia & Vetrò, 2020). Explainable artificial intelligence (XAI) is an emerging research field that develops techniques for making systems interpretable, transparent, and comprehensible (Doran et al., 2018; Futia & Vetrò, 2020; Gianfagna & Di Cecco, 2021). It would be great to be able to use XAI to understand how the country information interacts with the other variables in the input vector and ultimately contributes to the output. However, "a practical, widely applicable technology for explainable AI has not emerged yet" (Pedreschi et al., 2019, p. 9781). To the best of my knowledge, available model explainers (such as Biecek & Burzykowski, 2021) can only handle scalar, but not vector regression problems with more than one output variable. Hence, this appears to be a very promising research area for the years to come.

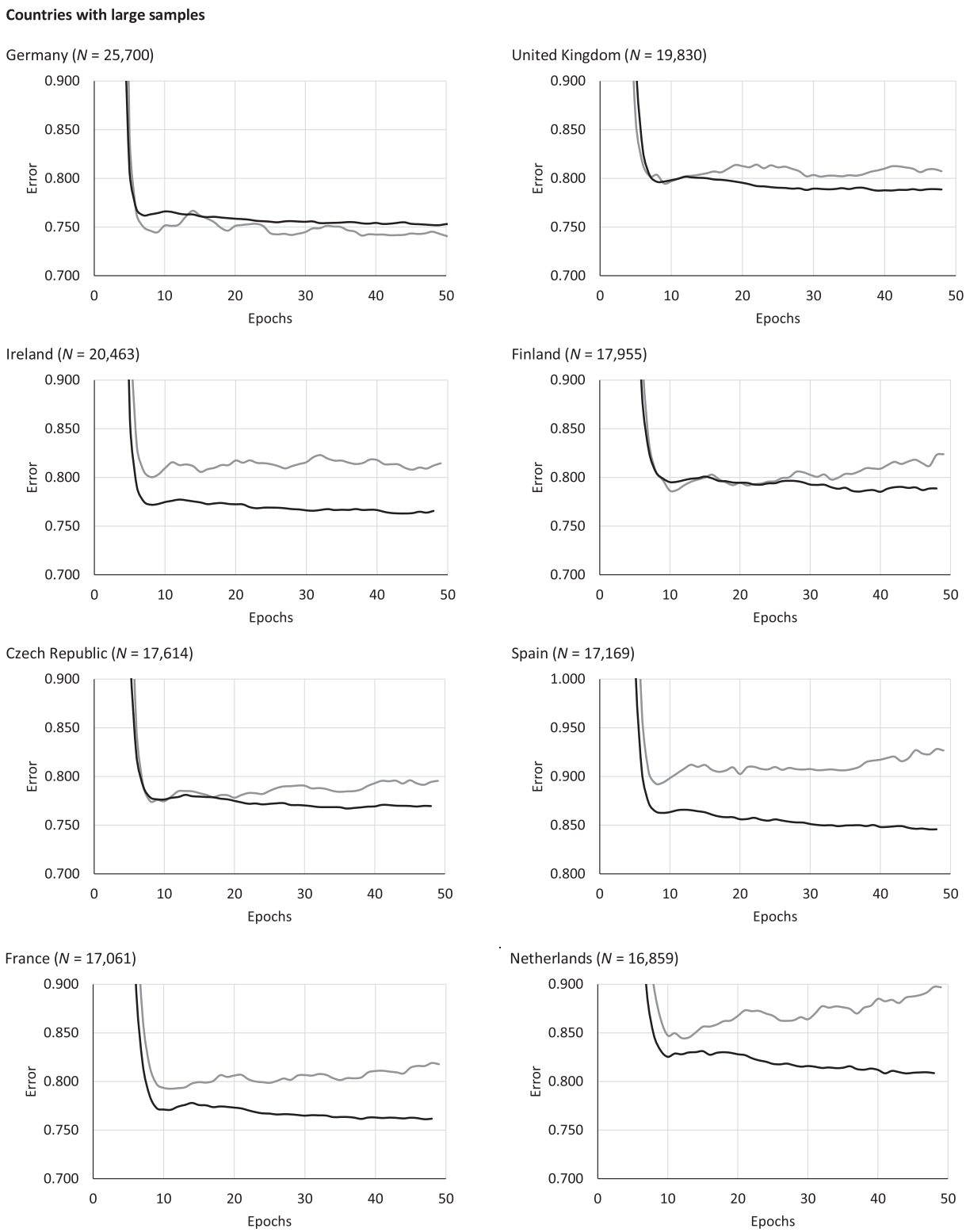
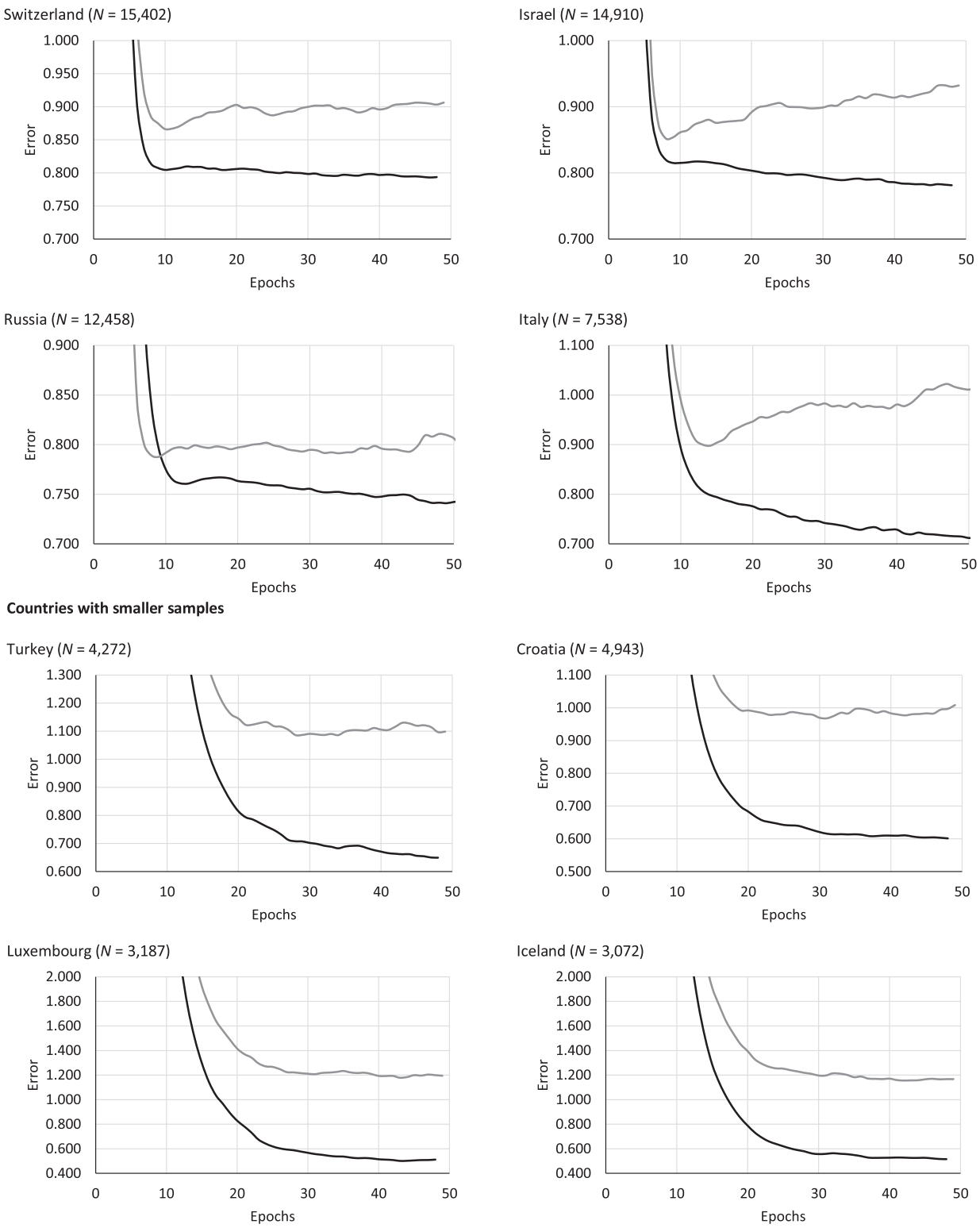


Fig. 1. Networks trained separately for each country.
The diagrams show changes in the interpolation (black line; training error) and generalization capabilities (grey line; generalization/testing error) during a training cycle of 50 epochs for 10 countries using network configuration A, as measured by a mean-squared-error loss function. A 60/40 split for training and testing data is used. Because of oscillations between epochs, a rolling average of five epochs is shown. Note that the scale of the error axis is different in some panels to better show the network's training behavior.

Countries with medium samples



Countries with smaller samples

Fig. 1. Continued

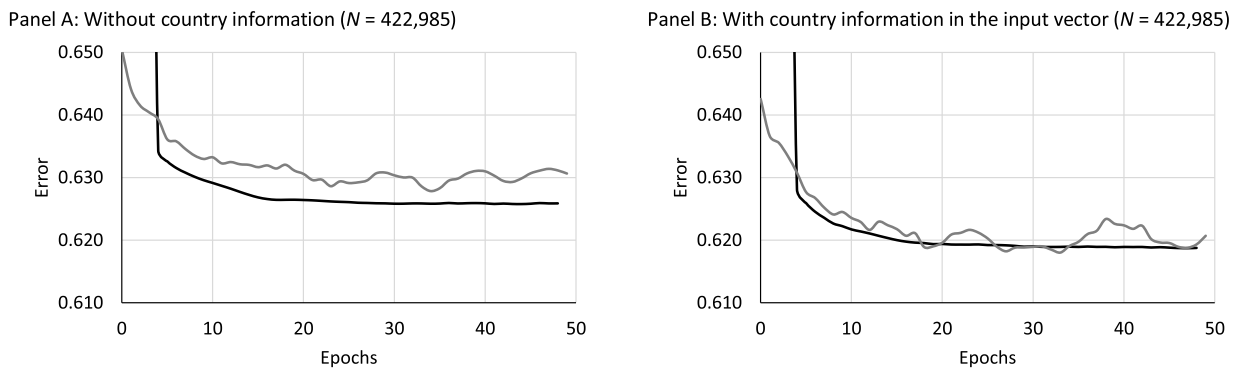


Fig. 2. Network trained with pooled countries.
Similar to Fig. 1, the diagrams show changes in the interpolation (black line; training error) and generalization capabilities (grey line; generalization/testing error) during a training cycle of 50 epochs, as measured by a mean-squared-error loss function. A 60/40 split for training and testing data is used. Contrary to Fig. 1, only one network is built with data from all 33 countries. In Panel A, only the original 159 variables are used to train the network (configuration A). In Panel B, country information is added to the input vector in the form of placeholder variables (coded as 0 and 1; configuration B). It is obvious that country information improves both the interpolation and generalization capabilities of a network: Both error curves in Panel B are notably lower than in Panel A.

Conclusion

As outlined at the outset of this article, when artificial intelligence applications are deployed around the world and interact in and with situations shaped by humans, cultural background information can be important for deciding on the best action. A deep artificial neural network, which has simple information about the society in which it operates, can abstract cultural values throughout its layers. This allows the system to better integrate itself into the human society. When "build[ing] machines that work in tight interdependence with people, […] engineers need to respect and learn from social scientists who are studying the complexities of human interaction with one another and with technology" (Cassell, 2019, p. 30). I, therefore, suggest designers of AI systems to experiment with this simple but effective form of machine enculturation proposed in this article.

The importance of cultural alignment cannot be overstated and is not an invention of AI scientists of our times. Aristotle, a philosopher and polymath during the classical period of ancient Greece, wrote already in 328 BC: "Anyone who cannot live in a community with others, or who does not need to because of his self-sufficiency, […] is either a wild beast or a god" (Aristotle & Reeve, 2017, p. 4).

When intelligent artifacts are developed for service robots, their communication and interaction process is not necessarily only an information exchange or a service delivery process, but it is a process embedded in the personal relationship between agents. For example, a service robot may need to recognize customers who want to be treated differently – maybe because they come from another country or have a different sociocultural background. Early on in AI research it was anticipated that "robots should be accepted by humans as communication partners. If people are used to and enjoy interacting to a human and not to a machine, they might not accept the robot" (Dautenhahn, 1995, pp. 352–353). The new field of machine enculturation can help intelligent artifacts integrate into the complex social system of the human society by giving them social expertise, so that they are perceived by humans as neither wild beasts – nor gods.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jjimei.2022.100118.

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